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Get to Know SAS® Studio

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Abstract

Attend this session for a point-and-click tour of SAS® Studio.

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Topics and Interface

Topics

Tools for Users Who Work with SAS Programs

1. Access a SAS Program
2. Run a SAS Program and Explore the Output
3. Rearrange the SAS Studio layout
4. Use the Format Code and Autocomplete Features
5. View and Export the Program Results and Summary
6. Create and Use Snippets

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Topics

Tools for Users Who Don't Work with SAS Programs

1. Build a SAS Studio Flow including:
   a. Import a File
   b. Query a Table
   c. Use a Step

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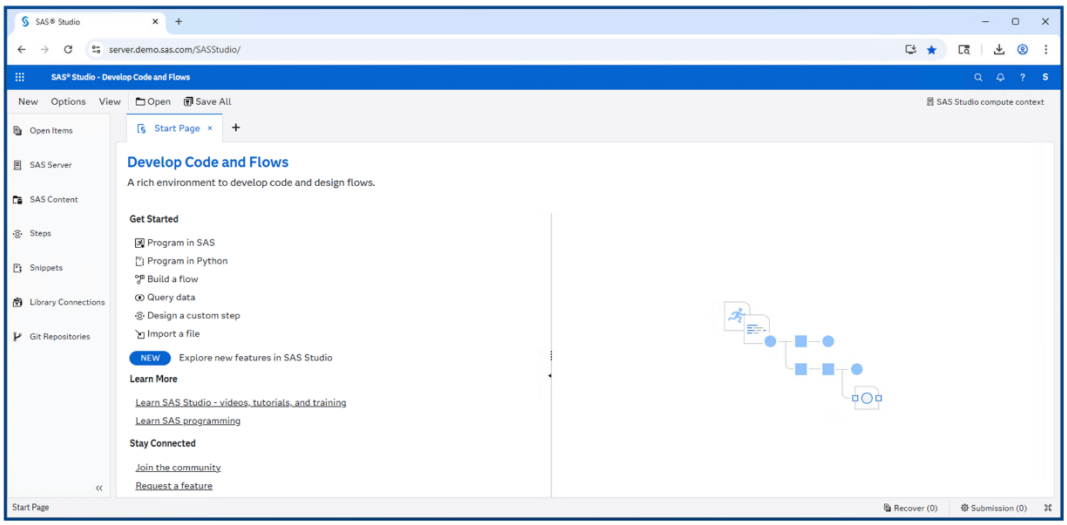
SAS Studio for SAS Viya



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Demo: Explore SAS Studio Features

Overview

In this demonstration, we explore various point-and-click tools in **SAS Studio** that help you write code and investigate your output.

*Note: If at any point you need to log into **SAS Viya**, use the following credentials:*

*User ID: **student***
*Password: **Metadata0***
*Assumable Groups: **No***

Access SAS Studio and Open a SAS Program

1. The machine should already be logged in to **SAS Viya** and opened to **SAS Landing**.
2. On the **SAS Viya** toolbar, click **⁝⁝⁝ (Applications menu)** and select **Develop Code and Flows**.
   This opens the **SAS Studio** application.
3. Select the **SAS Server** pane and navigate to **Files→Home→workshop→SIWSTD**.
4. Right-click the **SIWSTD** folder and select **Add folder shortcut**.
5. In the **SAS Server** pane, navigate to **Folder Shortcuts→My Shortcuts→Shortcut to SIWSTD** and double-click **sp01d01.sas** to open the file.
   Depending on the length of your path, a folder shortcut might make navigating to your files easier.

Run a SAS Program and Explore the Output

1. On the program toolbar, click **Run**.
   Notice the red X on the program tab indicating an error.
2. Click the **Log** tab to see a log summary indicating one warning and one error. Click each message in the log summary to highlight the message in the full log.
3. In the program **Code**, on **line 1**, change **daat** to **data**, and on **line 6**, change **fish** to **fish2**.
4. On the program toolbar, click **Run**.
5. Click the **Log** tab to confirm there are no warnings or errors, the **Results** tab to see a report, and the **Output Data** tab to see a table.
6. On the program toolbar, click **Save**.
7. On the **Output Data** tab, click the **Show or hide columns** icon.
8. Check **Select all** to deselect all columns. Then check **Species** and **Weight**.
9. Click outside the window to update the table.
   Notice the table now only shows 2 columns.
10. Click the **Use an expression to filter the table.** icon.

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11. In the **Expression Builder** window, on the **Data** tab, double-click **Species** to add it to the expression.
12. Click the **equals sign (=)** to add it to the expression.
13. On the **Values** tab, click **Get values**. Double-click **Pike** to add it to the expression.
    The final expression should look like this:
    ```
    Species= 'Pike'
    ```
14. Click **Validate** to validate the expression syntax.
15. Click **Save**.
    Notice the table now only displays 17 rows.
16. Right-click **Weight** and select **Sort** then select **Descending**.
17. On the program toolbar, click the **Submit the program in the background using another session.** icon.
    While the program runs in the background, you are free to run other programs in the current session. A notification will appear in the bottom right corner when the program is done running.
18. In the **SAS Server** pane, navigate to **Folder Shortcuts→My Shortcuts→Shortcut to SIWSTD**.
    Notice two new files, **sp01d01.html** and **sp01d01.log**. These were generated by the background submit.
    *Note: If you don't see the new files, right-click the **Shortcut to SIWSTD →Refresh**.*

Rearrange the SAS Studio Layout

1. Click and hold the **Log** tab, drag it to the left until you see a blue box appear, and release.
2. Click **X** on the program tab to close the file.
3. On the **SAS Viya** toolbar, click the **user icon (S)** and select **Settings**.
4. In the **Settings** window, select the **Code and Log** tab.
5. Use the drop-down to change the **Detail tab layout** from **Vertical** to **Standard**.
6. Click **Close**.
7. In the **SAS Server** pane, navigate to **Folder Shortcuts→My Shortcuts→Shortcut to SIWSTD** and double-click **sp01d02.sas** to open the file.
   Notice all tabs are together and the view is no longer split.

Use the Format Code and Autocomplete Features

1. On the program toolbar, click the **Format program** icon.
   Notice the program was automatically formatted.
2. In the program **Code**, on **line 17**, type **p**.
   Notice the autocomplete options appear. Select **proc**.
3. Use the autocomplete feature to enter the code below:
   ```
   proc print data=mpg_forecast;
   ```
4. On the **SAS Viya** toolbar, click the **user icon (S)** and select **Settings**.
5. In the **Settings** window, select the **Editors** tab.

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6. Click **Editor Settings**.
7. On the **Editing** tab, use the drop-down to change **Show auto-complete suggestions for other code** from **Suggestions list** to **Never**.
8. Click **Save** and then click **Close**.
9. On **line 18**, type **r**.
   Notice the autocomplete options no longer appear.
10. Complete the **RUN** statement to end the step:

```
proc print data=mpg_forecast;
run;
```

View and Export the Program Results and Summary

1. On the program toolbar, click **Run**.
2. On the **Results** tab, click **⋮ (More options)→Download**.
   Notice most of the options are grayed out. The two that are available are HTML (**Results file**) and text (**Listing file**).
3. On the **SAS Viya** toolbar, click the **user icon (S)** and select **Settings**.
4. In the **Settings** window, select the **Results** tab.
5. Check the box for **Produce PDF output**.
6. Click **Close**.
7. On the program toolbar, click **Run**.
8. On the **Results** tab, click **⋮ (More options)**, select **Download**, and select **PDF file**.
9. On the browser toolbar, click the downloaded file, **sp01d01-results.pdf**, to view the PDF.
10. Click **X** on the browser tab to close the PDF.
11. On the **Results** tab, click **⋮ (More options)**, select **Open in a browser tab** and select **Summary**.
12. Click **X** on the browser tab to close the summary.
13. Click **X** on the program tab to close the file. Click **Don't Save**.

Create and Use Snippets

1. On the application toolbar, click **New→SAS program**.
2. Select the **Snippets** pane and select the **SAS Snippets** tab.
3. Navigate to **Standard→Data** and click and hold **Import XLSX File**, drag it into the new program window, and release.
4. In the new program, on **line 6**, highlight everything inside the quotation marks.
5. Select the **SAS Server** pane and navigate to **Folder Shortcuts→My Shortcuts→Shortcut to SIWSTD**.
6. Right-click **Sales.xlsx** and select **Insert as path**.
7. In the new program, on **line 12**, insert the code shown below:

```
footnote "Generated on %sysfunc(datetime(),dateampm.)";
```

8. Highlight the code on **line 12**, and on the program toolbar, click **Snippet**.

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9. In the **New Snippet** window, complete the following fields:
   Name: **Footnote**
   Abbreviation: **FT**
10. Click **OK**.
11. In the new program, unhighlight the code on **line 12** and click **Run**.
12. Click **X** on the program tab to close the file. Click **Don't Save**.
13. In the **SAS Server** pane, navigate to **Folder Shortcuts→My Shortcuts→Shortcut to SIWSTD** and double-click **sp01d01.sas** to open the file.
14. Select the **Snippets** pane and select the **My Snippets** tab.
15. Click and hold **Footnote**, drag it into the program window, and release on **line 5**.
16. On the program toolbar, click **Undo**.
17. On **line 5**, type **@ft** and hit **Enter**.
18. Click **Run**.
19. Click **X** on the program tab to close the file. Click **Don't Save**.

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Demo: Build a SAS Studio Flow

Overview

In this demonstration, we build a **SAS Studio** flow that imports an **Excel** workbook, queries the data, and produces a bar chart.

*Note: If at any point you need to log into **SAS Viya**, use the following credentials:*

*User ID: **student***
*Password: **Metadata0***
*Assumable Groups: **No***

Import a File

1. On the application toolbar, click **New** and select **Flow**.
2. Select the **SAS Server** pane and navigate to **Files→Home→workshop→SIWSTD**.
3. Click and hold **Sales.xlsx**, drag it into the flow window, and release.
   Notice an **Import File** node was added to the flow.
4. In the flow, on the import **Options** tab, click **Analyze**.
5. Click the **Preview and Modify Data** tab.
6. On the flow toolbar, click **Run**.
   Notice the two output ports on the **Import File** node are now shaded, indicating two tables were created, one for each worksheet.
7. Click the **Submitted Code and Results** tab. Then click the **Output Data** tab.
   Notice the two tables were created with generic names in the **WORK** library.
8. Click the **Flow** tab.
9. Right-click the first output port on the **Import File** node and select **Add a table**.
10. On the **Table Properties** tab, complete the following fields:
    Library: **WORK**
    Table Name: **AU**
11. Right-click the second output port on the **Import File** node and select **Add a table**.
12. On the **Table Properties** tab, complete the following fields:
    Library: **WORK**
    Table Name: **US**
13. On the flow toolbar, click **Run**.
14. Click the **Submitted Code and Results** tab.
    Notice the two tables were created with custom names in the **WORK** library.

Query a Table

1. Click the **Flow** tab.
2. Right-click the **US** table node and select **Add a query**.

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3. On the query **Options** tab, in the **Columns** pane, expand **t1 (US)**.
4. Click and hold **t1 (US)**, drag it into the **Select** tab, and release.
5. Select the **Country** column and click the **Remove selected columns.** icon.
6. Click the **Filter** tab.
7. In the **Columns** pane, click and hold **Birth Date**, drag it into the **Filter** tab, and release.
8. Use the drop-down to change the **Condition** from **Missing** to **Less than**.
9. Click the **Use the quick filter to select a value.** icon.
10. In the **Add Filter** window, complete the following field:
    Value: **01JAN1950**
11. Click **Filter**.
12. Click the **Sort** tab.
13. In the **Columns** pane, click and hold **Birth Date**, drag it into the **Sort** tab, and release.
14. In the flow, right-click the output port on the **Query** node and select **Add a table**.
15. On the **Table Properties** tab, complete the following fields:
    Library: **WORK**
    Table Name: **PRE1950**
16. In the flow, right-click the **Query** node and select **Run node**.
    Notice the output port on the **Query** node is now shaded, indicating a table was created.
17. Click the **Submitted Code and Results** tab.
    Notice the new table has 8 columns, 8 rows, and is sorted by **Birth Date** in ascending order.
18. Click the **Flow** tab.

Use a Step

1. Select the **Steps** pane and select the **SAS Steps** tab.
2. Navigate to **Visualize Data** and double-click the **Bar Chart** step to add it to the flow.
3. In the flow, click and hold the **PRE1950** node, drag it to the **Bar Chart** node, and release. There should now be a link between the two nodes.
4. Click the **Bar Chart** node.
5. On the bar chart **Data** tab, click the **+** next to **Category**.
6. In the **Select a Column** window, select the **Job Title** column.
7. Click **OK**.
8. On the flow toolbar, click the **Arrange nodes.** icon.
9. On the flow toolbar, click **Run**.
10. Click the **Submitted Code and Results** tab, then click the **Submitted Code** tab to see the generated code, the **Log** tab to confirm there are no warnings or errors, the **Results** tab to see the bar chart, and the **Output Data** tab to see the three tables.
11. On the flow toolbar, click the **Save the flow.** icon.
12. In the **Save As** window, navigate to **SAS Server→Folder Shortcuts→My Shortcuts→Shortcut to SIWSTD**.
13. Complete the following field:
    Name: **Sales Flow**

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14. Click **Save**.
15. Click **X** on the flow tab to close the file.

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